

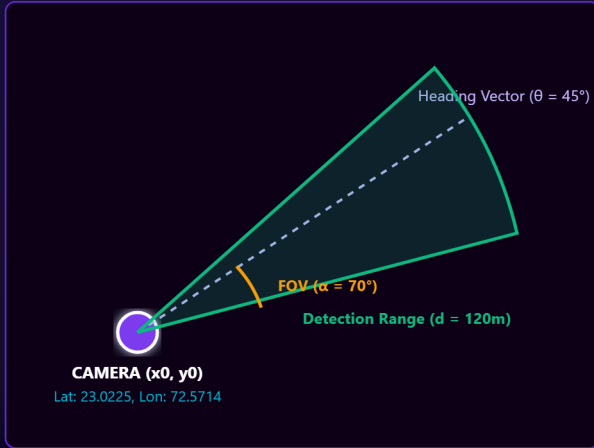


# PHANTOM 4.8 // DIAGRAM 07 — GIS SPATIAL INTELLIGENCE & CCTV COVERAGE WEDGES

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2  
EXHIBIT

## 1. CCTV COVERAGE WEDGE GEOMETRY & POSTGIS SPATIAL PROJECTION



### TRIGONOMETRIC FORMULATION

#### Coverage Wedge Apex Coordinates:

$$P1 = (x0 + d \cdot \sin(\theta - \alpha/2), y0 + d \cdot \cos(\theta - \alpha/2))$$

$$P2 = (x0 + d \cdot \sin(\theta + \alpha/2), y0 + d \cdot \cos(\theta + \alpha/2))$$

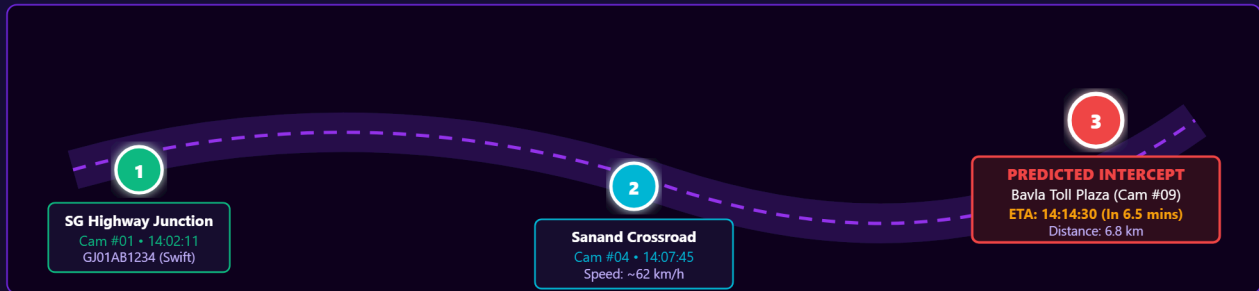
#### POSTGIS SPATIAL QUERY INTEGRATION:

- Stored as native PostGIS POLYGON geometry
- GIST Spatial Index on coverage\_polygon
- Queries: ST\_Within(vehicle\_pt, coverage\_wedge)

#### Automatic Blindspot Detection:

- ST\_Difference detects uncovered highway gaps
- Spatial Query Latency: < 1.9ms over 80K nodes

## 2. MULTI-CAMERA TRAJECTORY RECONSTRUCTION & INTERCEPT PREDICTION



### AUTOMATED POLICE DISPATCH & INTERCEPT PROTOCOL:

#### Toll Gate Auto-Barricade Pre-Alert

- Transmits alert payload to Bavla Toll Plaza C2 node
- Fast-lane boom barrier hold configured
- Target lane camera promoted to BURST\_TRACKING

#### Nearest PCR Interceptor Unit Alert

- Dispatches GPS coordinates to PCR Van #14 (2.1 km away)
- Live tactical HUD with vehicle color, make & plate
- Sub-second notification via Netram Mobile Gateway

## TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Calculates optical visual coverage wedge coordinates:  $(lat + d \cdot \sin \theta, lon + d \cdot \cos \theta)$  with azimuth and beam angle.
- Multi-camera trajectory engine connects sequential vehicle sightings along Gujarat National Highways (SG Highway -> Sanand).
- Predicts next intercept toll gate (e.g. Bavla Toll Plaza) with estimated arrival time (ETA) and auto-pre-alerts PCR units.